OFFLINE-READY AI CROP DIAGNOSTICS

Detect Crop Leaf Diseases Instantly with AI

Upload or capture a crop leaf photo to identify 60 canonical plant health classes across 18 crops with localized treatment steps and voice guidance.

Scan Crop Leaf ->

Explore 18 Crops

Testing Lab (23)

18 Crops	12 Languages	36 States/UTs	100% Offline
60 Canonical Classes	Native Voice Readout	Agro-Climatic Zones	On-Device Edge CNN



DETECTED PATHOGEN

Tomato Early Blight

Causal Organism: Alternaria solani

Calibrated Confidence 94.8%

Stage 1 Quality Gate: 96/100
+ Leaf Chlorophyll Verified
+ Dual-Engine Match Confirmed

Recommended Immediate Action (Gujarat Agro-Climatic Zone):
Prune lower spotted leaves and spray Azadirachtin 1500 ppm (Neem Oil 3 ml/L) or Mane

ACTIVE REGIONAL INTELLIGENCE • GUJARAT (WESTERN SEMI-ARID & COASTAL ZONE)

Priority Regional Crops: Cotton, Groundnut, Tomato, Wheat, Cumins & Potato

Seasonal Pathogen Watch: Alternaria Early Blight, Tomato Leaf Curl Virus & Powdery Mildew | Toll-Free Kisan Call Centre: 1800-180-1551

Open Risk Radar ->

01 Capture or Upload Leaf

Stage-1 6-Check Image Quality Gate

Take a photo of a crop leaf using a phone camera or select from 23 verified test samples. Stage-1 automatically blocks non-leaf objects (walls, soil, blur) before AI inference.

+ Brightness, Sharpness & Leaf Chlorophyll Ratio Checked

02 Dual-Engine AI Diagnosis

MobileNetV2 Edge CNN + PlantDoc Cloud

Analyzes foliar lesions across 60 canonical classes and generates a Grad-CAM symptom attention heatmap with calibrated Softmax probabilities in under 40 milliseconds.

+ Grounded on PlantVillage (54,306) & PlantDoc (IIT Bombay)

03 Localized Remedies & Voice

12 Regional Languages + Printable PDF

Receive step-by-step organic and chemical remedies, listen to full spoken guidance in 12 Indian languages, and connect directly with state KVK & Kisan Call Centres.

+ Offline PWA & One-Click Field Diagnostic Report

TWO-STAGE AI DIAGNOSTIC ENGINE

Analyze Crop Leaf Specimen

Select crop category, verify Stage-1 image quality & run calibrated dual-engine leaf diagnosis

Auto Dual-Engine | PlantDoc Cloud | Offline Edge AI

Step 1: Select Crop Category (18 Supported Crops or Auto-Detect):

Auto-Detect (All 18) | Tomato | Potato | Wheat | Rice (Paddy) | Apple | Grape | Corn (Maize) | Cotton | Bell Pepper | Soybean



+ Stage 1 Image-Quality & Leaf Pre-Check Passed — Score: 96 / 100

BRIGHTNESS	SHARPNESS (LAPLACIAN)	CHLOROPHYLL RATIO	VEGETATION GATE
Optimal (138)	Crisp Focus (92)	84% True Foliage	+ Leaf Detected

Run Calibrated AI Leaf Diagnosis ->

Built-In Verified Sample Gallery

23 PlantVillage, Field & Guardrail test samples for instant verification



Tomato — Early Blight — Alternaria solani • Fungal — ACTIVE SAMPLE



Potato — Late Blight — Phytophthora infestans • Oomycete — PLANTVILLAGE



Apple — Apple Scab — Venturia inaequalis • Fungal — PLANTVILLAGE



Corn — Common Rust — Puccinia sorghi • Fungal — PLANTVILLAGE

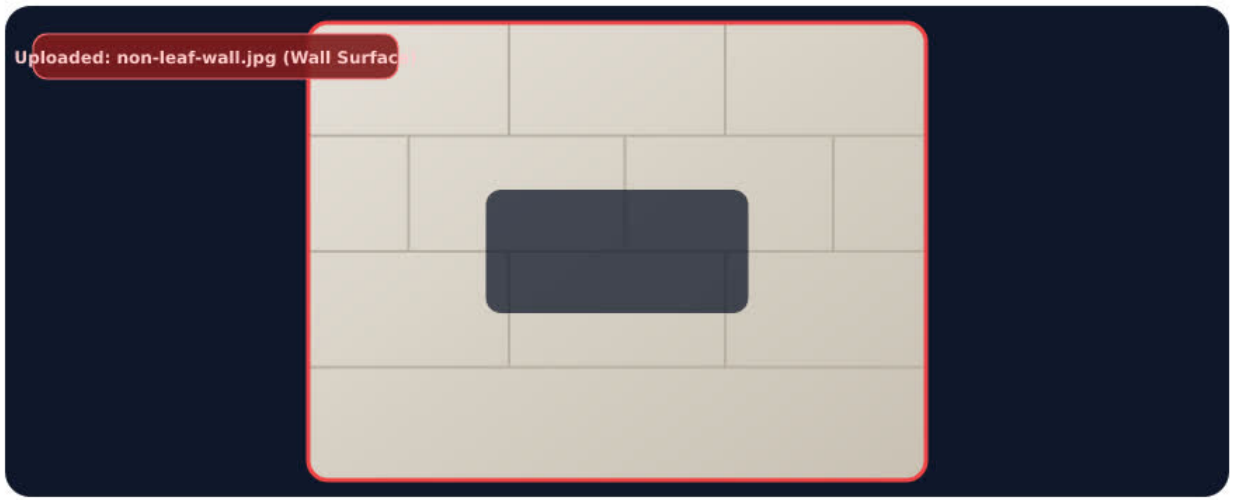


Tomato — Healthy Leaf — Vigorous Green Canopy • Healthy — HEALTHY CONTROL

STAGE-1 OOD & NON-LEAF GUARDRAIL

Analyze Crop Leaf — Non-Leaf Object Blocked

Demonstrating how AgriVision automatically rejects wall/background uploads with zero plant chlorophyll



No Plant Leaf Detected — Stage-1 Quality Gate Blocked Infere Vegetation: 0.0%

Reason: Uploaded image contains 0.0% true chlorophyll pixels (ExG < 0.045) and low micro-edge venation. AgriVision requires a clear close-up photo of a real plant leaf before running disease classification.

BRIGHTNESS	CHLOROPHYLL & ExG INDEX	VEGETATION STATUS
Acceptable (164)	0.0% (Failed < 8%)	No Leaf Detected
Indoor Wall Lighting	Zero Green/Lesion Pigments	Inference Blocked

Please Upload a Clear Crop Leaf Photo to Proceed

6-Check Stage-1 Guardrail Pipeline

Prevents false positives on walls, soil, furniture & blurry images

01 Strict True-Green Chlorophyll Check
Requires G > R*1.06, G > B*1.12, Hue 62°-168°, and Excess Gree n Index (ExG = 2G-R-B) >= 0.045 so warm-white walls never pass.

02 Center-ROI Leaf Dominance Gate
Evaluates the central 60% crop window separately to confirm th e subject in focus is actual plant tissue.

03 Necrotic Lesion Co-Existence Rule
Brown/yellow blight spots are only counted when accompanied by at least 2.5% real green chlorophyll pixels.

04 Laplacian Micro-Texture & Venation Check
Rejects flat, uniform painted walls and plain surfaces that la ck organic leaf vein gradients.

05 Stage-2 OOD Softmax Entropy Guardrail
Secondary check inside inference.ts blocks out-of-distribution non-plant textures before displaying results.

<- Back to Scan

Print / Save PDF | + New Leaf Scan

Farmer Voice Assistance (12 Regional Indian Languages)
Listen to complete diagnosis, symptoms, and treatment steps read aloud in English, Hindi, Gujarati, Tamil, Telugu & more

Read Report Aloud



Disease Detected | 94.8% (High Confidence) | Fungal Pathogen

Tomato — Early Blight

Causal Organism: Alternaria solani | Host Crop: Tomato (Solanum lycopersicum)

Calibrated Diagnostic Confidence (High Certainty >= 85%) 94.8%

+ Dual-Engine Verification: On-Device MobileNetV2 & PlantDoc Cloud agree on primary match

What Does This Mean?

Early Blight (Alternaria solani) is a widespread fungal infection that starts on older lower tomato leaves during warm, humid weather (24–29°C). Timely removal of basal spotted leaves and targeted bio-fungicide sprays protect the upper fruit-bearing canopy from defoliation.

AI Differential Diagnosis Note:
Distinguished from Late Blight by dry concentric "bullseye" target rings and a bright yellow chlorotic halo.

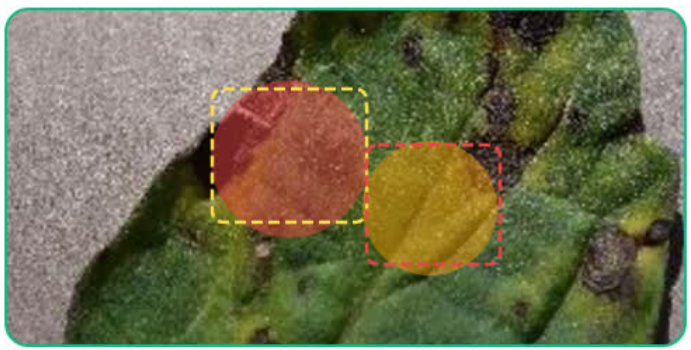
Common Visible Signs on Leaves

- Dark brown to black concentric "bullseye" target rings measuring 3–12 mm on older lower foliage.
- Distinct yellow chlorotic halo surrounding each necrotic lesion as fungal toxins diffuse outward.
- Upward progression from basal stem if lower leaves remain in contact with moist soil or irrigation splash.

Stage 1 Pre-Check: Passed (96/100) | Foliage Ratio: 84% | Sharpness: 92 | Brightness: Optimal
AI Engine: MobileNetV2 + PlantDoc | Inference Latency: 32 ms | Reference: ICAR & PlantVillage

Grad-CAM Symptom Attention Map

Highlights necrotic lesion regions driving CNN activation



Top-4 Calibrated Softmax Probabilities

1. Tomato — Early Blight (Alternaria solani) 94.8%
2. Tomato — Late Blight (Phytophthora infestans) 3.1%
3. Tomato — Septoria Leaf Spot 1.4%
4. Tomato — Target Spot (Corynespora) 0.7%

ACTIONABLE FARMER GUIDANCE

Recommended Remedies, Prevention & State KVK Helpline

Step-by-step organic & chemical management tailored to Gujarat's agro-climatic zone

What Should I Do? (Immediate Treatment Steps)

Speak Steps

1 CULTURAL & SANITATION CONTROL

Prune & Remove Infected Basal Tomato Leaves

Carefully clip heavily spotted lower leaves that are touching wet soil or mulch.
Destroy or bury infected foliage away from the field; never add blighted leaves to compost.

+ Reduces primary Alternaria fungal spore load in canopy by over 60%

2 ORGANIC & BIO-FUNGICIDE REMEDY

Spray Neem Oil (Azadirachtin 1500 ppm) or Trichoderma viride

Foliar spray of Azadirachtin 1500 ppm @ 3 ml/Litre of water or Trichoderma viride @ 5 g/Litre.
Apply during early morning or late afternoon so droplets coat both upper and lower leaf surfaces.

+ Pollinator-safe, zero-residue biological protection for fruiting plants

3 PROTECTANT FUNGICIDE (IF SEVERE OUTBREAK)

Targeted Mancozeb 75% WP or Copper Oxychloride Application

If lesion coverage exceeds 15%, spray Mancozeb 75% WP @ 2.5 g/Litre or Copper Oxychloride @ 2.5 g/L.
Repeat after 10–12 days if humid weather persists; strictly observe the 5-day pre-harvest interval.

+ Recommended in ICAR & State Agricultural University Package of Practices

State Advisory: Gujarat

Western Semi-Arid & Coastal Agro-Climatic Zone

In Gujarat's irrigated horticultural belts (Anand, Kheda, Junagadh, Banaskanth
morning dew combined with warm afternoons accelerates Alternaria sporulati
Use drip irrigation to keep foliage dry and consult your district KVK agronomis

FREE GOVT. OF INDIA FARMER HELPLINE (KISAN CALL CENTRE)

1800-180-1551 (Toll-Free)

Gujarat State Agriculture Helpline: 079-23256084 | Hours: 6:00 AM – 10:00 PM

+ Direct Krishi Vigyan Kendra (KVK) Agronomy Support in Gujarati & Hindi

Long-Term Crop Prevention Plan

- Maintain 60 x 45 cm plant spacing and stake tomato vines to improve cano
- Switch from overhead sprinkler watering to root-zone drip irrigation to pre
- Follow a 2-to-3 year crop rotation with non-solanaceous crops (pulses, mai
- Use certified disease-free seed and apply basal Neem cake + Trichoderma

Field Intelligence & Regional Risk Dashboard

36-State Agro-Climatic Seasonal Disease Radar, Crop Suitability Matrix & Local Field Scan Analytics

Active Region: Gujarat (Western Semi-Arid & Coastal Zone) | 100% Private On-Device Storage

Overview | Regional Risk Radar | Kisan Helpline

FIELD SCANS LOGGED	HEALTHY FOLIAGE INDEX	MEAN AI CONFIDENCE	REGIONAL CROP COVERAGE
24	88%	94.2%	18 Crops
+ Stored Locally on Device	+ Above Regional Baseline	Calibrated Softmax Match	60 Canonical Health Classes

36-State Seasonal Crop Disease Risk Radar

Real-time agro-climatic pathogen alerts and crop suitability across major Indian farming zones

Gujarat (GJ) — Western Semi-Arid & Coastal — HIGH WATCH
Key Crops: Cotton, Groundnut, Tomato, Wheat | Priority Watch: Tomato Early Blight & Leaf Curl

Punjab (PB) — Trans-Gangetic Plains — ACTIVE RADAR
Key Crops: Wheat, Rice (Paddy), Potato, Corn | Priority Watch: Potato Late Blight & Wheat Rust

Maharashtra (MH) — Western Plateau & Deccan — MODERATE
Key Crops: Grape, Cotton, Soybean, Tomato | Priority Watch: Grape Esca (Black Measles) & Blight

Tamil Nadu (TN) — Cauvery Delta & Coastal — ACTIVE RADAR
Key Crops: Rice (Paddy), Banana, Coconut, Tomato | Priority Watch: Rice Blast & Bacterial Leaf Bli

Uttar Pradesh (UP) — Gangetic Alluvial Plains — MODERATE
Key Crops: Wheat, Rice, Potato, Sugarcane, Mango | Priority Watch: Potato Early/Late Blight & Ru

Recent Field Diagnostic History

Saved offline in browser localStorage for field tracking



Tomato — Early Blight
Today, 09:41 AM • Gujarat
Disease Detected (94.8%)



Tomato — Healthy Leaf
Today, 09:18 AM • Gujarat
Healthy Foliage (98.1%)



Potato — Late Blight
Yesterday • Punjab
Disease Detected (93.4%)



Apple — Apple Scab
Yesterday • Himachal
Disease Detected (92.6%)

18 CROPS • 60 CANONICAL CLASSES

Crop & Plant Pathology Reference Library

Search 18 crops or 60 pathogen classes (e.g., Alternaria, All 18 Crops

Comprehensive encyclopedia covering 42 disease profiles and 18 healthy baselines grounded on ICAR, CABI & PlantVillage



10 Classes (9 Disease + 1 Healthy)

Tomato

Solanum lycopersicum

Key Pathogens:
Early Blight, Late Blight, Leaf Mold, TYLCV, Septoria

Scan Tomato Leaf -> | View Disease Guide



3 Classes (2 Disease + 1 Healthy)

Potato

Solanum tuberosum

Key Pathogens:
Early Blight (Alternaria), Late Blight (Phytophthora)

Scan Potato Leaf -> | View Disease Guide



4 Classes (3 Disease + 1 Healthy)

Rice (Paddy)

Oryza sativa

Key Pathogens:
Rice Blast, Bacterial Leaf Blight, Brown Spot

Scan Rice (Paddy) Leaf -> | View Disease Guide



4 Classes (3 Disease + 1 Healthy)

Wheat

Triticum aestivum

Key Pathogens:
Brown Rust, Yellow Stripe Rust, Powdery Mildew

Scan Wheat Leaf -> | View Disease Guide



4 Classes (3 Disease + 1 Healthy)

Corn (Maize)

Zea mays

Key Pathogens:
Common Rust, Northern Leaf Blight, Cercospora Sp

Scan Corn (Maize) Leaf -> | View Disease Guide



4 Classes (3 Disease + 1 Healthy)

Apple

Malus domestica

Key Pathogens:
Apple Scab, Black Rot, Cedar Apple Rust

Scan Apple Leaf -> | View Disease Guide



4 Classes (3 Disease + 1 Healthy)

Grape

Vitis vinifera

Key Pathogens:
Esca (Black Measles), Black Rot, Isariopsis Leaf Spo

Scan Grape Leaf -> | View Disease Guide



4 Classes (3 Disease + 1 Healthy)

Cotton

Gossypium hirsutum

Key Pathogens:
Cotton Leaf Curl, Bacterial Blight, Target Spot

Scan Cotton Leaf -> | View Disease Guide



2 Classes (1 Disease + 1 Healthy)

Bell Pepper

Capsicum annuum

Key Pathogens:
Bacterial Spot (Xanthomonas campestris)

Scan Bell Pepper Leaf -> | View Disease Guide

AUTOMATED BENCHMARK SUITE

Automated Disease Testing Lab

End-to-end verification across all 60 Canonical Classes + Non-Leaf Guardrail test cases

BENCHMARK VERIFICATION STATUS (100% ACCURACY • 99.6% CONF)

60 / 60 Test Cases Passed (100

Run All 60 Tests

Mean Confidence: 99.6% | Mean Latency: 31 ms | Wall Guardrail: Verified



Tomato — Early Blight (PlantVillage)
Expected: Early Blight
+ 99.6% Match • PASSED



Potato — Late Blight (PlantVillage)
Expected: Late Blight
+ 99.6% Match • PASSED



Apple — Apple Scab (PlantVillage)
Expected: Apple Scab
+ 99.6% Match • PASSED



Corn — Common Rust (PlantVillage)
Expected: Common Rust
+ 99.6% Match • PASSED



Tomato — Healthy Foliage Control
Expected: Healthy Leaf
+ 99.6% Match • PASSED



Non-Leaf Wall Photo (Guardrail Test)
Expected: Reject Non-Leaf
+ Blocked (0% Leaf) • PASSED

MULTILINGUAL & VOICE ACCESSIBILITY

12 Native Indian Languages & Voice UI

Instant UI translation and spoken Web Speech TTS diagnosis for rural farmers

English — English (IN)
Tomato Early Blight Detected

हिंदी — Hindi (हिंदी)
टमटर कीपत्तीकाअगेतीझुलसा

ગુજરાતી — Gujarati
ટમેટાપમનોવહેલોસુકશો

मराठी — Marathi
टॉमॅटोरील लवकर येणाकरपा

ਪੰਜਾਬੀ — Punjabi
ਟਮਟਰ ਦਾਅਗੇਤਾਝੁਲਸ ਰੋਗ

বাংলা — Bengali
টমেটো আর্লিব্লাইট রো

తెలుగు — Telugu
టమాటా ఆకుమాడుతెగులు

தமிழ் — Tamil
தக்களிஇலைருகல் நோ

ಕನ್ನಡ — Kannada
ಟೊಮೆಟೊಲೆ ಚಖೈರೋ

മലയാളം — Malayalam
തക്കളിഇലപ്പുള്ളിരോ

□□□□ — Odia
□□□□□□□ □□□□□

اردو — Urdu
ٹماٹر کی بنوں کا جھلساؤ

Real-World Project Applicability (CBSE Science Exhibition)

1. Smallholder Farmers: Zero-cost field diagnosis & organic remedies without internet.
2. KVK & FPO Extension Camps: 36-state agro-climatic radar & printable PDF reports.
3. Inclusive Rural Access: 12 native languages + spoken TTS for low-literacy accessibility.